



Can Machine Learning Improve Credit Default Prediction?
A Comparison of Classical and Tree-Based Methods

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Abstract

In the field of consumer finance, the ability to accurately assess credit default risk is paramount; any miscalculation can result in either significant capital losses or the unnecessary exclusion of creditworthy individuals. This study evaluates the potential of machine learning to enhance predictive accuracy beyond the capabilities of traditional statistical frameworks. Leveraging a comprehensive dataset from Kaggle, which tracks borrower demographics, income, and detailed repayment histories, I examine the performance of several distinct classification strategies.

The analysis compares a baseline benchmark against logistic regression, a single decision tree, and a random forest ensemble. Beyond standard metrics like ROC curves and AUC, the evaluation focuses heavily on the confusion matrix to dissect the trade-off between sensitivity and specificity- a balance that is critical for real-world risk management.

The findings demonstrate that tree-based ensemble methods offer a substantial leap in discriminatory power compared to classical approaches. Notably, the random forest model not only achieved the highest AUC but also showed a superior capacity for identifying high-risk borrowers who were otherwise missed by the logistic baseline. Furthermore, variable importance metrics highlight that a borrower's history of delinquency remains the most potent predictor of future default. While machine learning models often lack the transparency of linear regressions, these results suggest that the gain in predictive precision provides a compelling argument for their integration into modern lending workflows.

1. Introduction

The 2008 financial crisis served as a stark reminder of the fragility inherent in traditional credit scoring models. While logistic regression has long remained the industry standard due to its transparency and alignment with regulatory requirements, the modern landscape of consumer finance- characterized by high-volume, high-dimensional data- demands more robust analytical tools. This report explores whether the perceived "black box" of machine learning offers a sufficient performance gain to justify its complexity within the strictly regulated banking environment.

Historically, credit risk has been quantified using classical statistical techniques that prioritize interpretability. However, these models typically rely on rigid linearity assumptions, which often fail to capture the nuanced, non-linear dependencies present in large-scale datasets. In response, machine learning frameworks- particularly tree-based methods- have emerged as compelling alternatives. While a single decision tree provides an intuitive hierarchical structure for decision-making, ensemble methods like random forests further enhance predictive power by aggregating multiple trees to minimize variance. Yet, the adoption of these models is frequently hindered by a trade-off: as predictive accuracy increases, the direct interpretability of the model's logic tends to diminish.

The objective of this study is to determine whether machine learning can demonstrably improve credit default prediction over classical benchmarks. By utilizing a large-scale, real-world credit dataset, I compare the efficacy of logistic regression against decision trees and random forest classifiers. The evaluation moves beyond simple accuracy, focusing on discriminatory power and the practical implications for credit risk management, specifically regarding the identification of high-risk borrowers.

1.1 Goal of the report

The goal of this report is to evaluate whether machine learning methods can improve the prediction of credit default relative to classical statistical models. In particular, the study aims to compare the predictive performance of logistic regression with tree-based methods, including a decision tree and a random forest classifier.

The analysis seeks to answer the following research questions:

1. How accurately can credit default be predicted using classical statistical models?
2. Do tree-based machine learning methods outperform logistic regression in terms of predictive performance?
3. Which borrower characteristics are the most important predictors of credit default?

By addressing these questions, the report aims to assess the usefulness of machine learning approaches in credit risk assessment and to highlight the trade-offs between predictive accuracy and model interpretability.

2. Data

2.1 Data Source

The analysis is based on the *Give Me Some Credit* dataset, which is publicly available on the Kaggle platform. The dataset was originally created for a credit risk prediction competition and contains detailed information on individual borrowers. It represents a realistic consumer credit dataset and is well suited for evaluating both classical statistical models and machine learning classification methods.

The dataset consists of 150 000 observations and 11 variables, including one binary target variable indicating whether a borrower experienced serious delinquency within a two-year period.

2.2 Data Overview

The target variable, *SeriousDlqin2yrs*, takes the value 1 if the borrower defaulted and 0 otherwise. The remaining variables describe borrowers' demographic characteristics, income, debt structure, and repayment history. These include measures of revolving credit utilization, age, monthly income, number of open credit lines and loans, number of real estate loans, number of dependents, and several indicators capturing past delinquency behaviour, such as the number of times a borrower was 30-59, 60-89, or 90 or more days past due.

Exploratory analysis revealed that the dataset is characterized by class imbalance, as only a small proportion of borrowers experienced serious delinquency. Such imbalance reflects real-world credit portfolios and poses a challenge for classification models. Several variables also exhibit highly skewed distributions and extreme values, particularly those related to debt ratios and credit utilization.

2.3 Data Cleaning

The dataset was processed using R to ensure consistency across models and to prepare the data for statistical analysis. A significant challenge during the preprocessing stage was the high level of noise in variables like *DebtRatio* and *MonthlyIncome*. Initial exploration revealed several impossible values (e.g., debt ratios exceeding 10 000%), which necessitated capping at the 99th percentile to prevent the logistic regression from failing to converge. For missing values, median imputation was chosen over simple deletion to preserve the sample size of 150 000 observations.

In addition, several variables contained extreme outliers that could disproportionately influence model estimation, especially in logistic regression and decision trees. To mitigate this issue, numeric variables were capped at the 1st and 99th percentiles. The target variable was converted into a categorical factor prior to model estimation.

2.4 Data Splitting

To evaluate model performance on unseen data, the dataset was randomly divided into a training set and a test set. Seventy percent of the observations were assigned to the training set, while the remaining thirty percent formed the test set. This train-test split was applied consistently across all models and served as the basis for model comparison and performance evaluation.

3. Methodology

This study applies several classification models to predict credit default and compares their performance using standard evaluation metrics. The analysis follows a structured modelling pipeline, beginning with a simple benchmark model and gradually increasing model complexity. All models are trained on the same training dataset and evaluated on the same test dataset to ensure comparability.

3.1 Benchmark Model

As a baseline for comparison, a benchmark model was constructed by assigning all observations the same predicted probability of default equal to the mean default rate observed in the training set. This approach represents random classification without the use of explanatory variables and serves as a reference point for evaluating the added value of more sophisticated models. The benchmark model is expected to achieve an area under the ROC curve (AUC) close to 0.5.

3.2 Logistic Regression

Logistic regression was used as the primary classical statistical model. The model estimates the probability of credit default as a function of borrower characteristics using a logistic link function. All explanatory variables were included in the model to assess their joint impact on default risk.

Logistic regression is widely applied in credit risk modelling due to its interpretability and theoretical transparency. Model coefficients can be directly interpreted in terms of the direction and magnitude of their association with default probability. However, logistic regression assumes linear relationships between predictors and the log-odds of default, which may limit its predictive performance in complex datasets.

3.3 Decision Tree

A decision tree classifier was estimated to capture non-linear relationships and interactions between explanatory variables. The tree recursively partitions the feature space into homogeneous regions by selecting splits that maximize class separation. To prevent overfitting, the complexity of the tree was controlled by limiting its maximum depth, specifying a minimum number of observations required for a split, and applying a cost-complexity pruning parameter.

Decision trees provide an intuitive representation of decision rules and are highly interpretable. However, single trees often suffer from high variance and may exhibit weaker predictive performance compared to ensemble methods.

3.4 Random Forest

To improve predictive accuracy and reduce model variance, a random forest classifier was employed. The random forest model constructs an ensemble of decision trees using bootstrap samples of the training data and random subsets of predictors at each split. Final predictions are obtained by aggregating the predictions across all trees.

In addition, class weights were introduced to address the imbalance between default and non-default observations, placing greater emphasis on correctly identifying default cases. Variable importance measures derived from the random forest were used to assess the relative contribution of individual predictors to model performance.

3.5 Model Evaluation

Model performance was evaluated using the test dataset, which was not used during model training. Predictive accuracy was assessed using receiver operating characteristic (ROC) curves and the area under the ROC curve (AUC), which measures a model's ability to discriminate between defaulters and non-defaulters across different classification thresholds.

In addition, confusion matrices were computed to examine classification performance in more detail. These matrices report the number of true positives, true negatives, false positives, and false negatives and allow for the calculation of accuracy, sensitivity, specificity, and precision. Particular attention was paid to sensitivity, as correctly identifying high-risk borrowers is of central importance in credit risk management.

4. Findings

4.1 Model Performance Comparison

The comparison of model performance reveals substantial differences between the considered approaches. As expected, the benchmark model performs no better than random classification, achieving an AUC of approximately 0.5. Logistic regression improves predictive performance considerably, reaching an AUC of around 0.70. This indicates that borrower characteristics and repayment history contain meaningful information for predicting credit default.

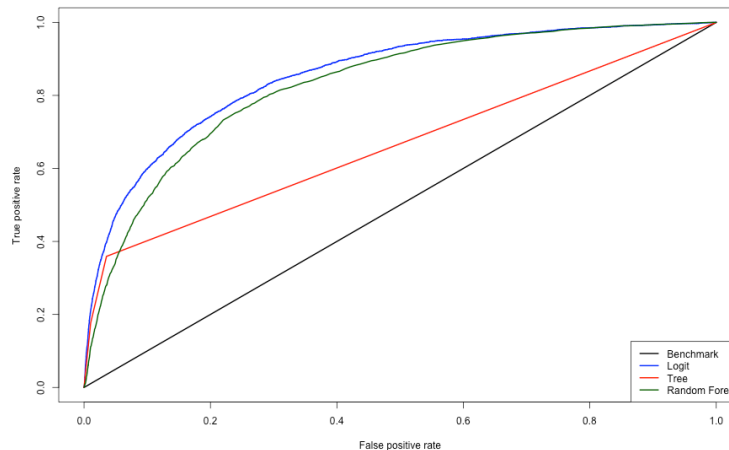


Figure 1: Receiver operating characteristic (ROC) curves for the benchmark model, logistic regression, decision tree, and random forest. The random forest model exhibits the strongest discriminatory power across all classification thresholds.

The decision tree model achieves a slightly lower AUC than logistic regression, suggesting that a single tree may not fully capture the underlying structure of the data. In contrast, the random forest model substantially outperforms all other approaches, achieving an AUC close to 0.85. This result highlights the advantage of ensemble learning in handling complex, non-linear relationships and interactions among predictors.

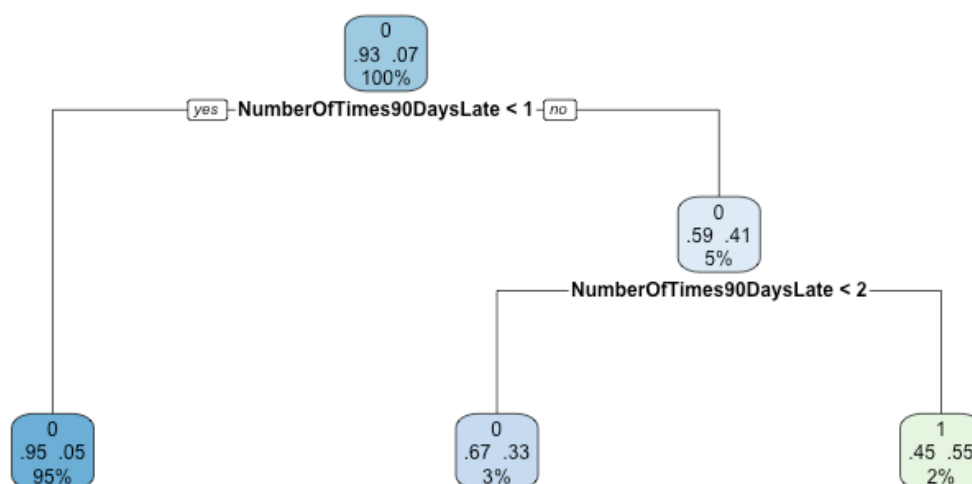


Figure 2: Decision tree visualization for credit default prediction. The model primarily relies on severe past delinquency measures, particularly the number of times a borrower was 90 or more days past due.

4.2 Variable Importance

Variable importance analysis provides insight into the main drivers of model predictions.

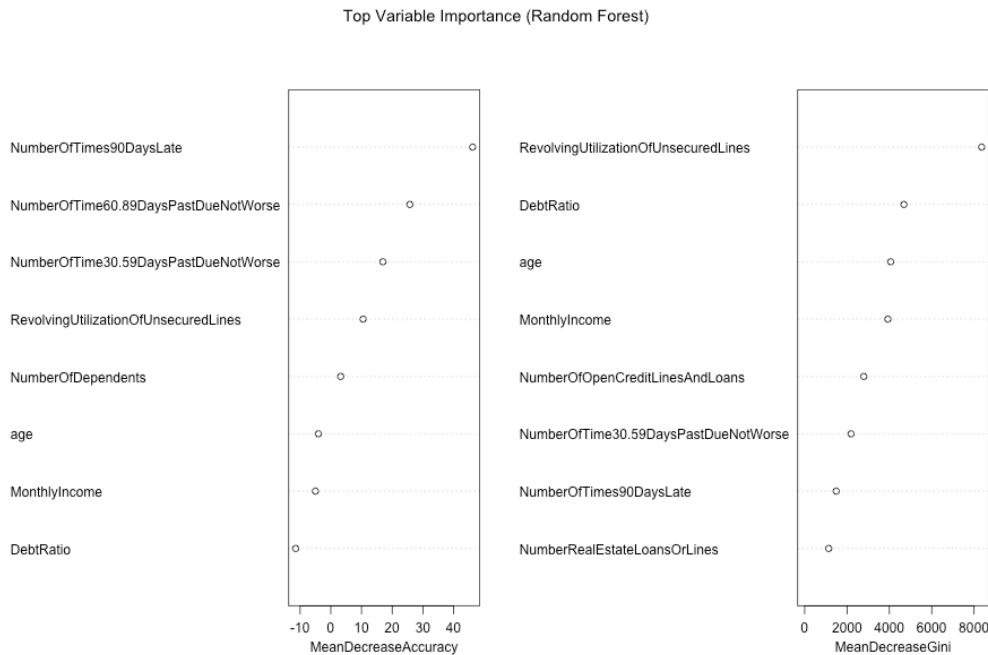


Figure 3: *Variable importance derived from the random forest model. Past delinquency indicators and credit utilization measures are the most influential predictors of credit default.*

Variable importance analysis based on the random forest model shows that past delinquency measures are the most influential predictors of credit default. Variables capturing the number of times a borrower was severely late on payments, particularly those related to delays of 90 days or more, dominate the importance ranking. Measures of credit utilization and debt burden also contribute significantly to model performance, while demographic variables such as age and number of dependents play a comparatively smaller role.

These findings are consistent with economic intuition and standard credit risk theory, which emphasize the importance of repayment history as a key indicator of future default behaviour.

4.3 Classification Performance

Confusion matrix analysis reveals notable differences in the classification behaviour of the models. Logistic regression exhibits high specificity, correctly identifying the majority of non-defaulting borrowers, but shows relatively low sensitivity, failing to detect a substantial proportion of defaulters. This conservative behaviour may reduce false positives but increases the risk of misclassifying high-risk borrowers as creditworthy.

	Reference		
Prediction		0	1
	0	41 076	2 071
	1	911	942

Table 1: *Confusion matrix for the logistic regression model evaluated on the test dataset. The model exhibits high specificity but relatively low sensitivity, indicating conservative classification behaviour. Note: 0 = non-default, 1 = default.*

The random forest model achieves higher sensitivity, identifying a larger share of defaulters at the cost of an increased number of false positives. From a practical perspective, this trade-off may be desirable in credit risk applications where the cost of granting credit to a high-risk borrower outweighs the cost of rejecting a low-risk applicant.

	Reference		
Prediction		0	1
	0	38 344	1 566
	1	3 643	1 447

Table 2: *Confusion matrix for the random forest model evaluated on the test dataset. The model achieves higher sensitivity at the cost of increased false positives, making it suitable for identifying high-risk borrowers. Note: 0 = non-default, 1 = default.*

The confusion matrix results are consistent with the ROC analysis, confirming the superior discriminatory performance of the random forest model.

5. Discussion

The findings of this research underscore that the selection of a predictive framework is not merely a technical choice but a strategic one in the context of credit risk. While traditional models like logistic regression are lauded for their transparency, their predictive ceiling becomes apparent when compared to the flexibility of machine learning. Specifically, the results confirm that tree-based ensembles are far better equipped to navigate the non-linearities and high-dimensional interactions inherent in modern consumer credit data.

Logistic regression proved to be highly effective at identifying reliable borrowers, as evidenced by its high specificity. This conservative "safety-first" behaviour is consistent with the transparency mandates of financial regulators. However, its diagnostic weakness lies in its low sensitivity; the model frequently fails to flag high-risk individuals. From an institutional standpoint, this lack of predictive depth could lead to an unacceptable accumulation of default losses that a more nuanced model might have avoided.

While single decision trees offer a visually intuitive map of risk- clearly identifying past delinquency as a primary red flag- their standalone utility is hindered by structural instability and weaker discriminatory power. In contrast, the random forest model emerged as the most robust predictor, achieving the highest AUC and a significantly improved sensitivity. By aggregating multiple trees, this ensemble approach successfully isolated the most critical predictors, such as repayment history, aligning with core tenets of credit risk theory. The trade-off, however, remains the "opacity" of the model, which poses a significant

hurdle in environments where every loan rejection must be explained to both the customer and the regulator.

Several caveats to these findings warrant mention. The reliance on a single train-test split means the results are subject to the specific variance of this sample. Furthermore, the predictive accuracy might be enhanced by including missing socio-economic indicators, such as employment duration or the total length of the borrower's credit history. Finally, without external validation on an independent dataset, these conclusions remain tied to the specific characteristics of the Kaggle data.

In conclusion, this study illustrates the central tension between model interpretability and raw accuracy. While classical models remain indispensable for regulatory compliance, machine learning offers a powerful toolkit for reducing default exposure, provided it is implemented with a clear understanding of its inherent complexity.

6. Conclusion

This research set out to determine whether the integration of machine learning can substantively enhance credit default prediction compared to established statistical methodologies. By evaluating a baseline benchmark, logistic regression, and tree-based models against a large-scale consumer dataset, the study highlights a clear hierarchy in predictive performance.

The empirical evidence confirms that ensemble learning- specifically the random forest classifier- significantly exceeds the discriminatory power of classical logistic regression. A key finding of this analysis is the dominant role of past delinquency measures in predicting future default, which reaffirms the enduring importance of repayment history in risk assessment. While logistic regression remains a robust tool for identifying low-risk applicants, its conservative nature results in a critical failure to flag a substantial portion of high-risk borrowers.

These results suggest that while machine learning models are inherently more complex, their ability to pinpoint high-risk cases offers significant economic value for financial institutions. However, the transition toward these "black box" models must be handled with caution, ensuring that gains in accuracy do not come at the expense of regulatory transparency or ethical compliance.

Looking ahead, the robustness of these findings could be further strengthened through the use of k-fold cross-validation, the exploration of gradient-boosted machines, or the inclusion of more granular borrower socio-economics. Ultimately, this study demonstrates that while they may not entirely replace traditional methods, tree-based machine learning models serve as a powerful and necessary complement to the modern credit risk modelling toolkit.